Photoacoustic 3D brain phantom and multi speed of sound reconstruction

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***Abstract*—Photoacoustic tomography (PAT) is emerging as a popular non-radioactive imaging modality in cerebral imaging research. However, skull-induced aberrations pose significant challenges for achieving clear cerebral vascular distribution in PAT cerebral imaging. Additionally, the limitation of available photoacoustic data hampers the application of data-driven correction methods. In this study, we developed a comprehensive three-dimensional numerical human brain phantom dataset, derived from clinical T1/T2-weighted magnetic resonance imaging (MRI) data and corresponding magnetic resonance angiography (MRA) data. A realistic and feasible experimental setup for Monte Carlo-based optical simulation and k-wave-based acoustic simulation was designed to generate photoacoustic data. We employed multi-speed-of-sound PAT image reconstruction using two conventional methods to test cerebral imaging. The evaluation results indicate that the reconstructed image quality is significantly enhanced through multi-speed-of-sound image reconstruction.**

Keywords—Photoacoustic tomography, human brain imaging, photoacoustic phantom simulation

I. Introduction

Photoacoustic tomography (PAT) is an emerging imaging modality that offers high optical contrast and deep penetration depth, making it a promising non-invasive technique for human cerebral imaging [1]. Recent studies indicate that photoacoustic tomography imaging can delineate the cerebral vascular distribution, which may prove beneficial in diagnosing cerebral diseases. Liming Nie et al demonstrated the feasibility for light to pass through entire adult human skulls using a photon recycler [2]. However, the skull-induced aberration presents significant challenges for human brain PAT imaging, particularly in achieving precise localization of cerebral vessels, especially within microvascular structures. The superimposition of newly generated side-lobe signals and the main lobe in Mie scattering mode significantly complicates the transcranial PAT image reconstruction process. This complexity arises because the acoustic wavelength of the photoacoustic signals generated by micro vessels is of the same order as the size of skull pores [3]. One approach to correct distortion in cerebral PAT image reconstruction is to incorporate additional information about the skull from other modalities. Chao Huang et al. proposed combining PAT with data on skull composition and morphology obtained from adjunct X-ray CT imaging to correct aberrations [4]. Also, precise modeling can aid in reconstructing more accurate PAT cerebral images. Shuai Na et al. proposed a three-layer back-projection image reconstruction model that considers the heterogeneity properties of the human skull [5]. Another approach to addressing the issue is through model-based image reconstruction methods [6]. Given an adequate amount of data, deep-learning-based methods can also effectively alleviate the skull-induced aberration [7].

In this work, we propose to generate a three-dimensional human brain numerical phantom dataset from clinical human brain T1/T2 weighted MRI and MRA data in IXI-dataset (https://brain-development.org/ixi-dataset/) for simulation and cerebral imaging research. Additionally, we propose a realistic and feasible optical and acoustic simulation experiment setup for enhancing photoacoustic data. Finally, we performed image reconstruction using two conventional methods with varying speed-of-sound map to determine whether PAT human brain imaging can provide an accurate cerebral vascular distribution. The results demonstrate that reconstructed image quality is significantly improved through multi-speed-of-sound image reconstruction.

II. Methods

A. Real 3D digital human brain phantoms generation

We propose to generate three-dimensional human brain numerical phantoms from clinical MRI and MRA data which accurately emphasize distinct anatomical structures and clear vascular distributions.

One of the primary challenges in photoacoustic tomography of cerebral imaging is achieving a clear visualization of cerebral vascular distributions. To construct

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elaborate numerical human brain phantoms that better mimic a real human brain anatomy, we utilize both T1/T2 weighted MRI data for high-contrast images of fundamental anatomical structures and corresponding MRA data for clearer vascular distribution. To obtain elaborate numerical human brain phantoms that can better mimic real human brain, we take both T1/T2 weighted MRI data with higher contrast of basic anatomy structure, alongside corresponding MRA data that offer clear vascular distributions. Initially, linear and non-linear image registration techniques are applied to ensure consistent sizing and precise localization of cerebral vessels. The SynthSeg [8] is employed to segment the registered MRI volumes into 8 distinct tissue types that are scalp, compact bone, spongy bone, cerebrospinal fluid, grey matter, white matter, eyeball, and muscle. For registered MRA data, we choose 3D Frangi filter [9] to segment vessel from MRA volume of the corresponding MRI samples. Further digital image processing techniques, including image erosion and connected component analysis, are applied to the filtered vascular data to enhance clarity. By integrating the segmented fundamental structures of human brain anatomy and vascular distributions, we generate comprehensive three-dimensional numerical phantoms of the human brain that each one is with voxel size of 256×256×130 and spatial resolution of 1 mm. The whole three-dimensional numerical human brain phantom generation process is illustrated in Fig. 1.

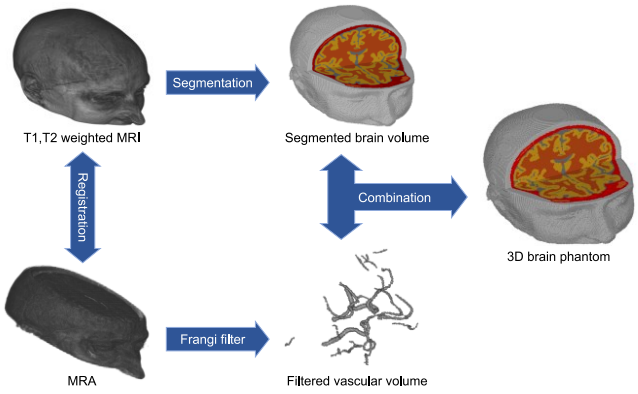


Fig. 1. Three-dimensional numerical human brain phantom generation process.

B. Optical simulation

Aiming at obtaining photoacoustic signals used for human brain PAT cerebral imaging, we separately perform optical and acoustic simulations on the generated numerical human brain phantoms.

In optical simulation, three-dimensional Monte Carlo method implemented with MCXLAB [10] is employed to model the photon propagation process within numerical human brain phantoms whose optical fluence distribution is determined by

$$F(\vec{r},t)=\frac{E_a P(\vec{r},t)}{E_t\sum_i\sum_j P(\vec{r_i},t_j)\mu_a(\vec{r_i})\Delta V\Delta t} \quad (1)$$

where $F(\vec{r},t)$ and $P(\vec{r},t)$ are the optical fluence and raw probability at position $\vec{r}$ and time t. ΔV is the voxel volume and Δt is the time-gate length. $\mu_a(\vec{r_i})$ is the medium absorption coefficient at position $\vec{r_i}$. E_a/E_t indicates the total energy absorbed by the corresponding medium. Optical parameters (optical absorption coefficients, reduced scattering coefficient, anisotropy and refractive index) of human brain tissues involved are given from literature [11]. Collimated Gaussian beam are chosen to be the light sources for simulation. A total of nine light sources are employed, eight of which are placed around the numerical phantom and one is placed at the top. The 8 Gaussian beam sources arranged in a ring formation with a radius of 128 mm and a vertical depth of 100 mm that centered at the central point of the axial slice of the brain phantom under the same depth. The configuration of ring formation light sources is illustrated in Fig. 2. The 8 horizontal Gaussian beams are with a waist radius of 60 mm while the top Gaussian beam is with a waist radius of 120 mm. The number of photon packages is set to one billion to simulate the photons' propagation inside the human brain numerical phantoms illuminated by the pulsed laser. The duration of the light pulses is five nanosecond with the time-gate is set to one nanoseconds. Quintuple down sampling of each voxel's size is employed to ensure comprehensive brain illumination.

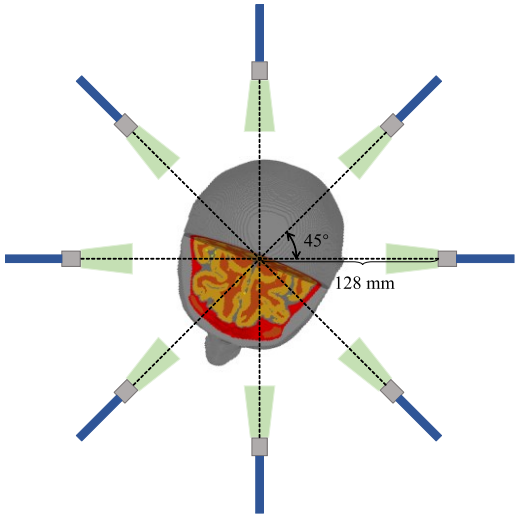


Fig. 2. Planform visualization of eight ring-formed Gaussian beams in optical simulation.

According to the thermoelastic theory [12] we can determine the initial pressure distribution $p_0(\vec{r})$ at position $\vec{r}$ can be derived by

$$p_0(\vec{r})=\Gamma\eta_{th}\mu_a(\vec{r})F(\vec{r}) \quad (2)$$

where $F(\vec{r})$ and $\mu_a(\vec{r})$ represent obtained optical fluence and tissue absorption coefficient at $\vec{r}$. Γ is Grueneisen parameter while η_{th} is the percentage of energy that is converted into heat, both of which remain constant at a given temperature. Consequently, the acquired optical fluence maps are multiplied by the optical absorption coefficients of the corresponding tissues to derive initial acoustic pressure distribution, which is then used for the subsequent acoustic simulation.

C. Acoustic simulation

The acoustic simulation is conducted using k-Wave toolbox in MATLAB [13]. The acoustic properties (speed of sound, density, attenuation coefficient) of brain tissues are set according to the measured value [14]. Since the initial pressure distribution does not entirely occupy the full 130 mm space in the perpendicular direction, we only consider an 80 mm vertical space containing valid values. Consequently, a 256×256×120 k-grid is defined as the region of interest for the acoustic simulation. Each cell is with a size of 1 mm × 1 mm

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× 1 mm. 2048-element cylindrical transducer array that composed of 16 layers of 128-element point-like ultrasound transducers ring array with a radius of 125 mm are used to receive the acoustic signals. The cylindrical array with a 6 mm interval between each layer simulates a plane scan of a 128-element ring array moving along the perpendicular direction. Motion distortion during the scan is neglected in this simulation. Each transducer possesses an 80 percent bandwidth and a central frequency of 1 MHz. The obtained initial pressure distribution is positioned at the center of the cylindrical array, and the sampling frequency is set to 12.5 MHz. The acoustic simulation setup is illustrated in Fig. 3.

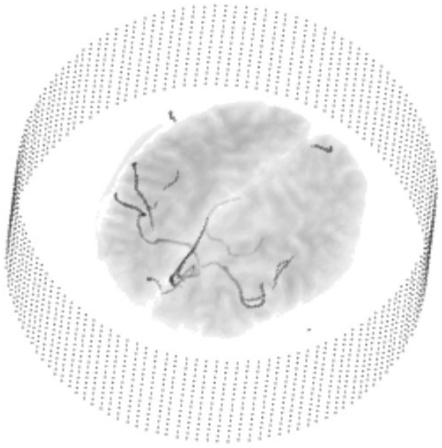


Fig. 3. A simple illustration of acoustic simulation setup.

The PA sinograms without aberration are considered reference data for quantitative evaluation. The reference data is obtained by manually setting acoustic properties of skull and scalp in numerical human brain phantoms to be the same as the coupling fluid during acoustic simulation.

D. Multi-speed-of-sound image reconsturcion

Three-dimensional delay-and sum (DAS) beamforming algorithm and time reversal (TR) algorithm are utilized for PAT image reconstruction in this work. According to acoustic signal propagation process, we can determine the initial pressure $p_0(\vec{r})$ at position $\vec{r}$ though [15]

$$(\nabla^2 - \frac{1}{v(\vec{r})}\frac{\partial^2}{\partial t^2})p(\vec{r},t) = -p_0(\vec{r})\frac{d\delta(t)}{dt} \qquad (3)$$

where $v(\vec{r})$ is the speed of sound at position $\vec{r}$ and $p(\vec{r},t)$ is the PA signal at position $\vec{r}$ and time t. With accurate speed of sound distribution we can derive the initial pressure distribution p_0 from the received PA signal.

III. Experiments and Results

In this work, we utilize the entire IXI dataset to generate numerical phantoms. After manual selection, a total of 560 three-dimensional numerical human brain phantoms are generated from real clinical brain data and subsequently used for photoacoustic simulations. Both homogeneous and heterogenous speed-of-sound maps are employed in PAT image reconstruction.

Axial, coronal and sagittal slices of a normalized human cerebral PAT volume reconstruction sample are presented in Fig. 3. The results demonstrate that time-reversal image reconstructions with heterogeneous speed-of-sound maps exhibit significantly improved performance compared to homogeneous reconstructions using either DAS or TR methods. However, the multi-speed-of-sound time-reversal reconstructed cerebral vascular distribution in the axial slice yields a lower contrast compared to the other two methods. Additionally, all three methods show significant distortions in the sagittal slices.

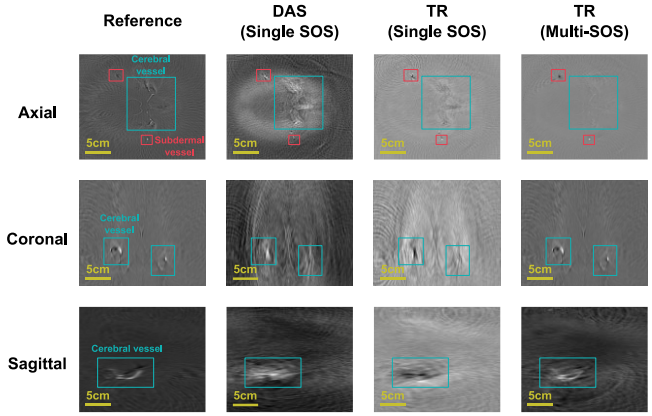


Fig. 4. Three views of slices from one sample's PAT volume reconstruction using different image reconstruction algorithms and speed-of-sound maps.

Sstructural similarity (SSIM) and peak signal-to-noise ratio (PSNR) of the normalized PA sinograms and normalized grayscale reconstructed PAT volumes are chosen as the quantitative evaluation metrics to access the performance of DAS and TR reconstruction methods using different speed-of-sound maps. Table I illustrates that both SSIM and PSNR are improved with multi-speed-of-sound image reconstruction while SSIM exhibits a more substantial improvement.

TABLE I. Quantitative Comparison of Results

	SSIM	**PSNR**
PA sinogram	0.807	26.943
DAS (Single speed-of-sound)	0.609	19.742
TR (Single speed-of-sound)	0.631	11.284
TR (Multi-speed-of-sound)	0.789	20.719

IV. Discussion and Conclusion

In this paper, we constructed a comprehensive three-dimensional numerical human brain phantom dataset derived from clinical human brain data, and proposed a realistic and accessible setup for three-dimensional optical and acoustic simulations. The dataset comprises a total of 560 numerical phantoms, which were used to evaluate the performance of PAT cerebral image reconstruction using DAS and TR algorithms with varying speed-of-sound maps. The results demonstrate that multi-speed-of-sound PAT image reconstruction can effectively delineate cerebral vascular structures and mitigate distortions caused by skull-induced aberrations.

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